Integer Programming

CS5491: Artificial Intelligence
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Content Credits: Prof. Wei's CS4486 Course and Prof. Boddeti's AI Course

TODAY

Formulate decisions with integer and binary variables.

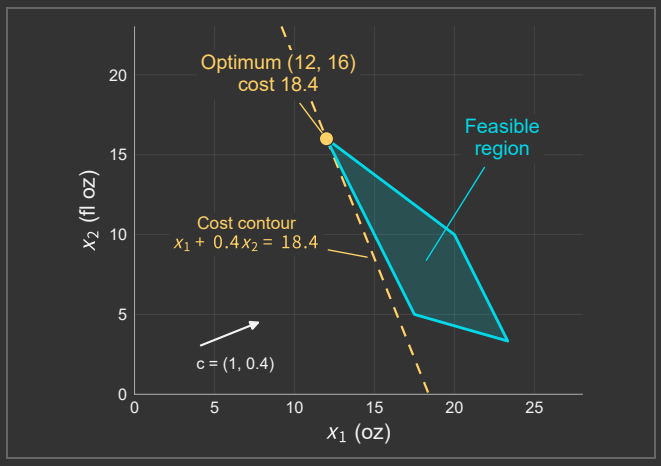
Use an LP relaxation to bound an integer program.

Solve an integer program by branch and bound.

LP RECAP: THE DIET PROBLEM

x_1: stir-fry (oz); x_2: boba (fl oz).

$$\begin{aligned} \min\quad &x_1+0.4x_2\\ \text{s.t.}\quad &2000\le100x_1+50x_2\le2500,\\ &3x_1+4x_2\le100,\\ &20x_1+70x_2\ge700,\\ &x_1,x_2\ge0. \end{aligned}$$



Unique optimum: $(12,16)$.
Minimum cost: 18.4.

FINDING A VERTEX

Choose n independent constraint boundaries for n variables.

Diet example

Calorie minimum and sugar limit:

$$\begin{aligned}100x_1+50x_2&=2000,\\3x_1+4x_2&=100.\end{aligned}$$

Solving gives $(x_1,x_2)=(12,16)$.

Notation for $A\mathbf{x}\le\mathbf{b}$

I: indices of the selected constraints.

A_I: selected rows of A.

$\mathbf{b}_I$: matching entries of $\mathbf{b}$.

Set these constraints to equality:

$$A_I\mathbf{x}=\mathbf{b}_I.$$

Check **all original constraints** before accepting the point as a vertex.

DISCRETE DECISIONS

Decision	Variable	Domain
Number of fixed 8 oz servings	n	$\mathbb{Z}_{\ge0}$
Whether to select an item	z	$\{0,1\}$
Amount of a divisible resource	t	$\mathbb{R}_{\ge0}$

Define the unit before writing the model.
For n servings of 8 oz stir-fry: amount $=8n$ oz, cost $=8n$, calories $=800n$.

LP, ILP AND MILP

Linear objective and linear constraints:
$$\min_{\mathbf{x}}\ \mathbf{c}^T\mathbf{x}\qquad\text{subject to }A\mathbf{x}\le\mathbf{b}.$$

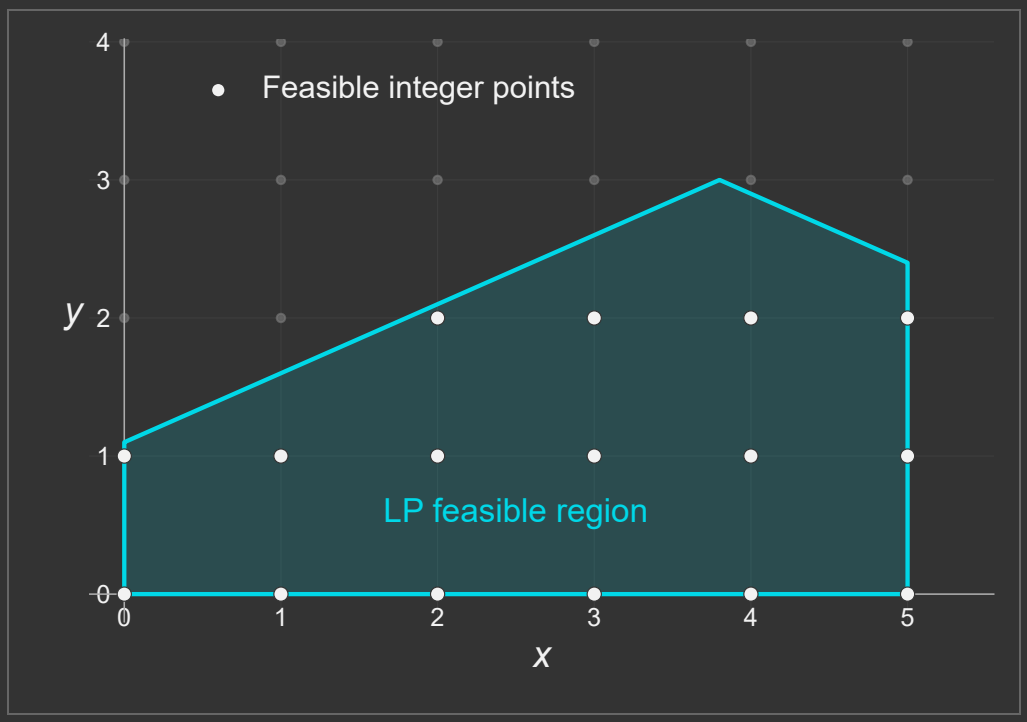
Model	Variable domains
Linear program (LP)	$x_i\in\mathbb{R}$ for all i
Pure integer linear program (ILP)	$x_i\in\mathbb{Z}$ for all i
Binary integer program	$x_i\in\{0,1\}$ for all i
Mixed-integer linear program (MILP)	Some variables integer; others continuous

INTEGER PROGRAMMING EXAMPLE

$$\begin{aligned} \min_{x,y}\quad &z=x-4y\\ \text{subject to}\quad &-10x+20y\le22,\\ &5x+10y\le49,\\ &0\le x\le5,\quad y\ge0,\\ &x,y\in\mathbb{Z}. \end{aligned}$$

We will use this model for relaxation, rounding and branching.

INTEGER FEASIBLE POINTS



LP: every point in the shaded region. IP: only the integer points inside it.

LP RELAXATION

Integer program

$$\begin{aligned}\min\quad&\mathbf{c}^T\mathbf{x}\\\text{s.t.}\quad&\mathbf{A}\mathbf{x}\le\mathbf{b}\\&\mathbf{x}\in\mathbb{Z}^n\end{aligned}$$

LP relaxation

$$\begin{aligned}\min\quad&\mathbf{c}^T\mathbf{x}\\\text{s.t.}\quad&\mathbf{A}\mathbf{x}\le\mathbf{b}\\&\mathbf{x}\in\mathbb{R}^n\end{aligned}$$

Drop integrality. Keep the objective, constraints and variable bounds.

WHY RELAXATION GIVES A BOUND

Every integer feasible point is also LP feasible:
$$P_{\mathrm{IP}}\subseteq P_{\mathrm{LP}}.$$
Minimizing over a larger set cannot increase the optimum:
$$z^*_{\mathrm{LP}}\le z^*_{\mathrm{IP}}.$$
The LP optimum is a **lower bound** for minimization.

CONNECTION TO INFORMED SEARCH

In informed search, solving a relaxed problem can give an optimistic cost estimate.

At a branch-and-bound node, the LP relaxation bounds the best integer solution allowed by that node.

This bound helps rule out branches that cannot improve the current solution.

A* uses $g(n)+h(n)$ to estimate total cost. Here, the node LP bounds the **total objective directly**.

CHECK YOUR UNDERSTANDING

For a minimization IP and its LP relaxation with finite optima:

Statement

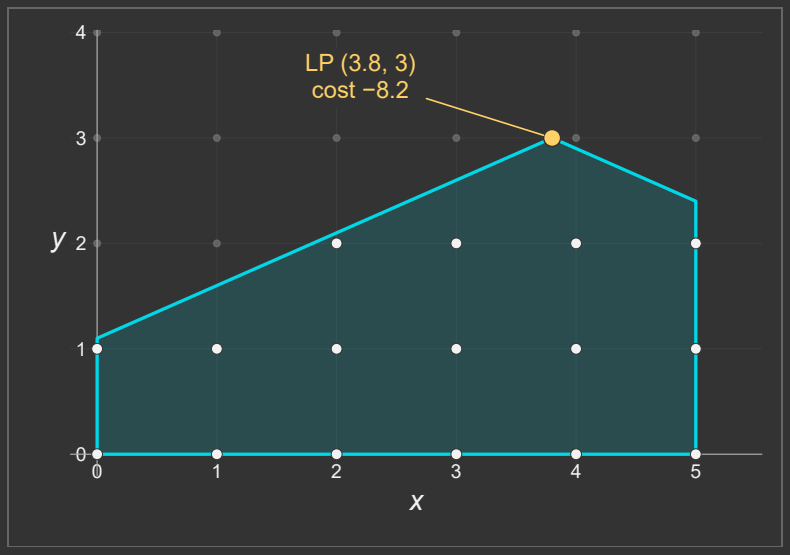
$z^*_{\mathrm{LP}}\le z^*_{\mathrm{IP}}$

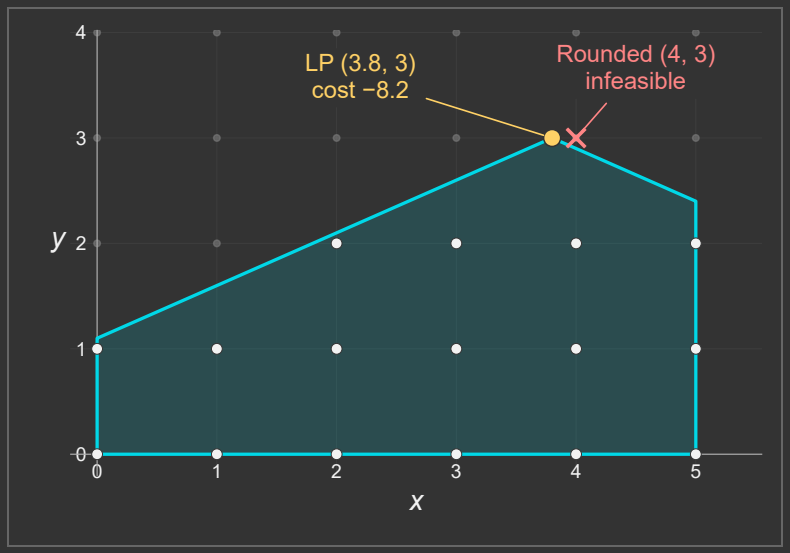
$\mathbf{x}^*_{\mathrm{LP}}=\mathbf{x}^*_{\mathrm{IP}}$

An LP-optimal point with integer coordinates solves the IP.

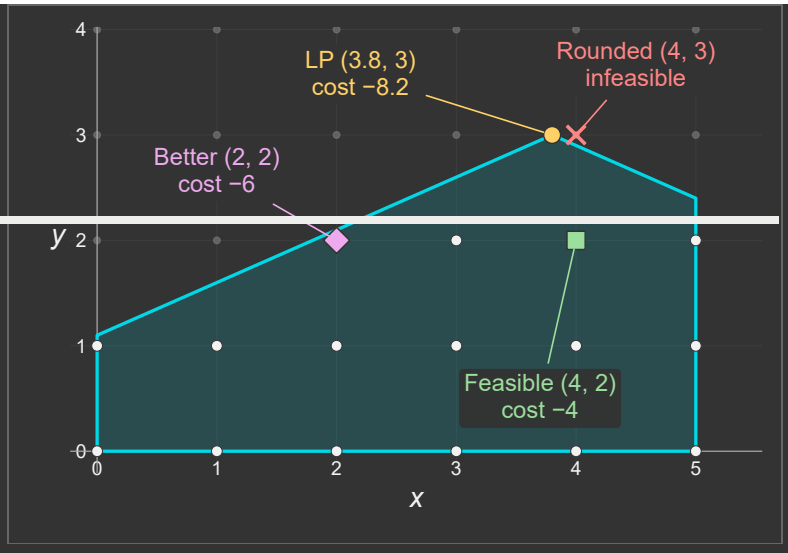
WHY ROUNDING CAN FAIL

LP optimum: $(3.8,3)$, value -8.2.
Nearest rounding gives $(4,3)$:
$5(4)+10(3)=50>49$.
Infeasible.
Nearby feasible $(4,2)$ costs -4, but $(2,2)$ costs -6.
A nearby feasible point need not be optimal.





BRANCH AND BOUND



Incumbent: the best feasible integer solution found so far.

1. Solve the LP relaxation at the current node.
2. Close the node if it is infeasible, integral, or cannot improve the incumbent.
3. Otherwise, branch on a fractional integer variable.
4. Explore the children, keeping all ancestor constraints.

When no open nodes remain, the best integer solution found is optimal.

THE BRANCHING RULE

Choose an integer-constrained variable with fractional value x_i^*.

$$x_i\le\lfloor x_i^*\rfloor\qquad\text{or}\qquad x_i\ge\lceil x_i^*\rceil.$$

Example: $x_i^*=3.8$.

$$x_i\le3\qquad\text{or}\qquad x_i\ge4.$$

Both children exclude the current fractional solution.

Every integer feasible solution belongs to one of the children.

BOUNDS AND THE INCUMBENT

Node lower bound L: the optimal value of its LP relaxation.

Incumbent: the best feasible integer solution found so far.

Global upper bound U: the incumbent's objective value.

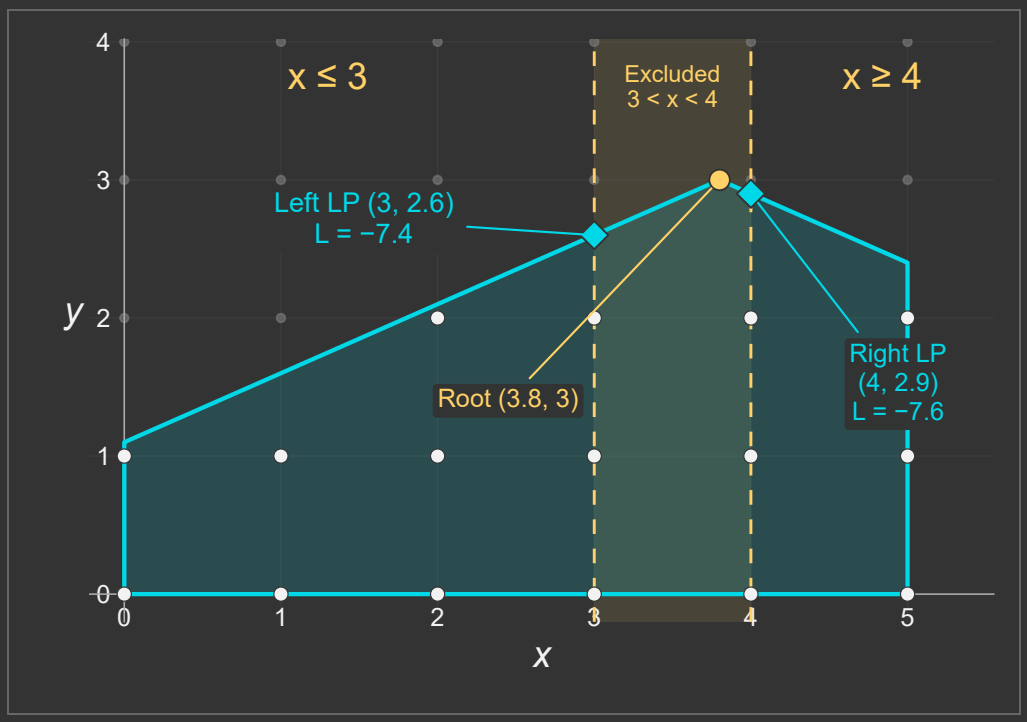
Start with $U=+\infty$ if no feasible integer solution is known.
When a better integer solution of value z is found, set $U\leftarrow z$ and store that solution.

THREE WAYS TO CLOSE A NODE

Case	Action and reason
LP is infeasible	No integer solution can satisfy this node's constraints.
LP optimum is integral	Update the incumbent if better. This node is solved.
Bound: $L\ge U$	No integer solution in this node can improve the incumbent.

Example: $L=-5.4$ and $U=-6$. Prune because $-5.4\ge-6$.

EXAMPLE: THE FIRST BRANCH



Root: $(3.8,3)$, $L=-8.2$. Branch on x and explore $x\ge4$ first.

THE FIRST INCUMBENT

Within $x\ge4$, the LP optimum is $(4,2.9)$ with $L=-7.6$. Branch on y: $y\le2$ or $y\ge3$.

Additional constraint	LP result	Action
$y\le2$	$(4,2)$, value -4	Integer: set $U=-4$.
$y\ge3$	Infeasible	Close this node.

The $x\ge4$ subtree is finished. Return to $x\le3$.

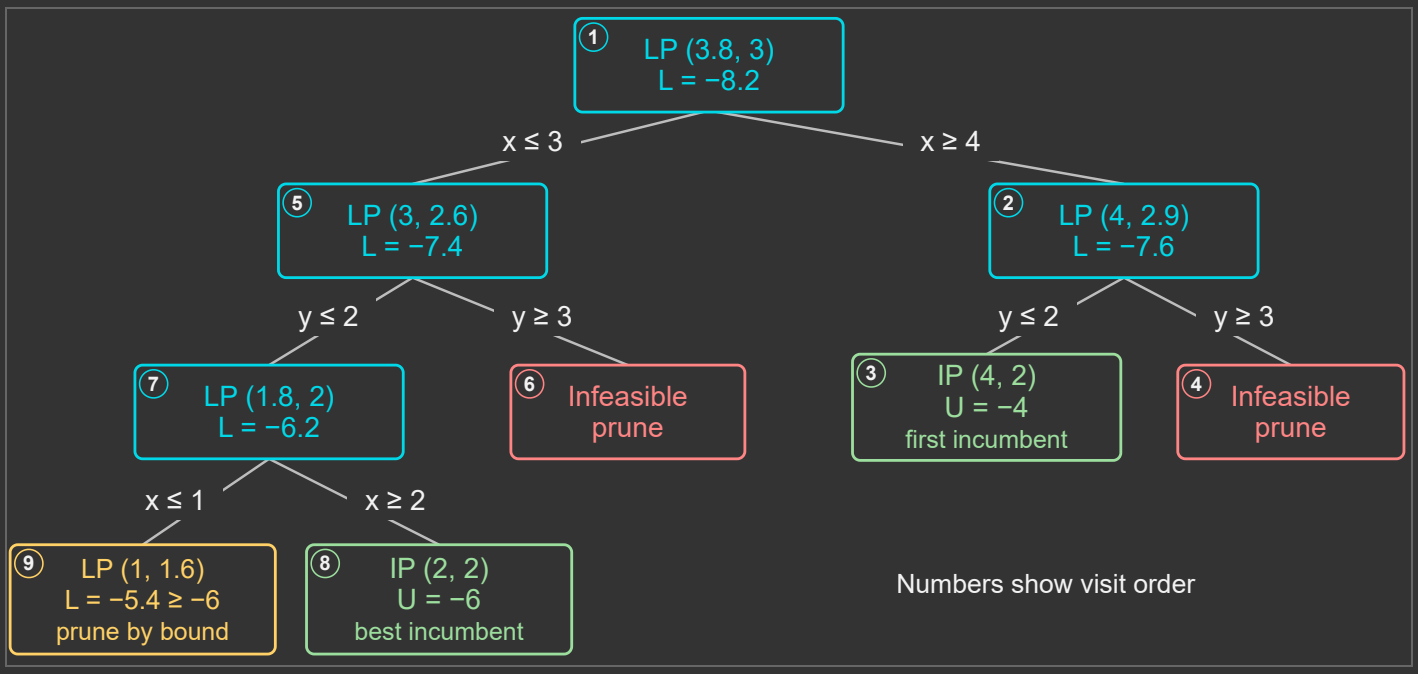
IMPROVING THE INCUMBENT

Within $x\le3$: LP optimum $(3,2.6)$, $L=-7.4$. Entering this subtree: $U=-4$.

Additional constraints	LP result	Action
$y\ge3$	Infeasible	Close.
$y\le2$	$(1.8,2)$; $L=-6.2$	Branch on x.
$y\le2,\ x\ge2$	$(2,2)$; value -6	Integer: update $U=-6$.
$y\le2,\ x\le1$	$(1,1.6)$; $L=-5.4$	Prune: $-5.4\ge-6$.

Every row inherits $x\le3$ and the original constraints.

THE COMPLETE SEARCH TREE



Incumbent: $U=+\infty\ \longrightarrow\ -4\ \longrightarrow\ -6$. Every leaf is closed.

OPTIMALITY CERTIFICATE

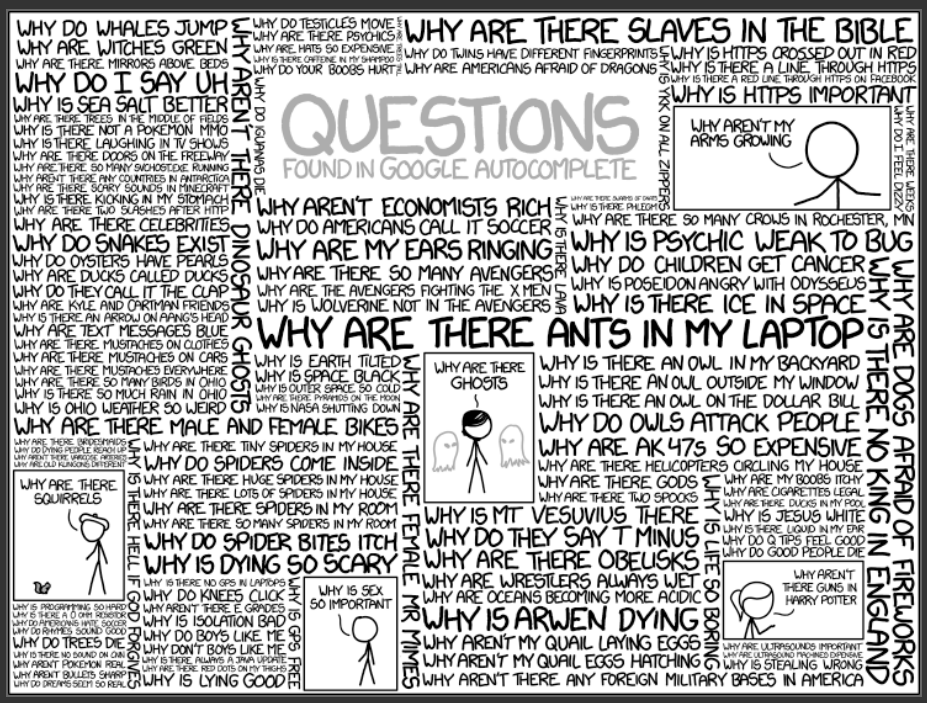
$$\mathbf{x}^*=(2,2),\qquad z^*_{\mathrm{IP}}=2-4(2)=-6.$$

Every leaf is infeasible, solved integrally, or unable to improve -6.

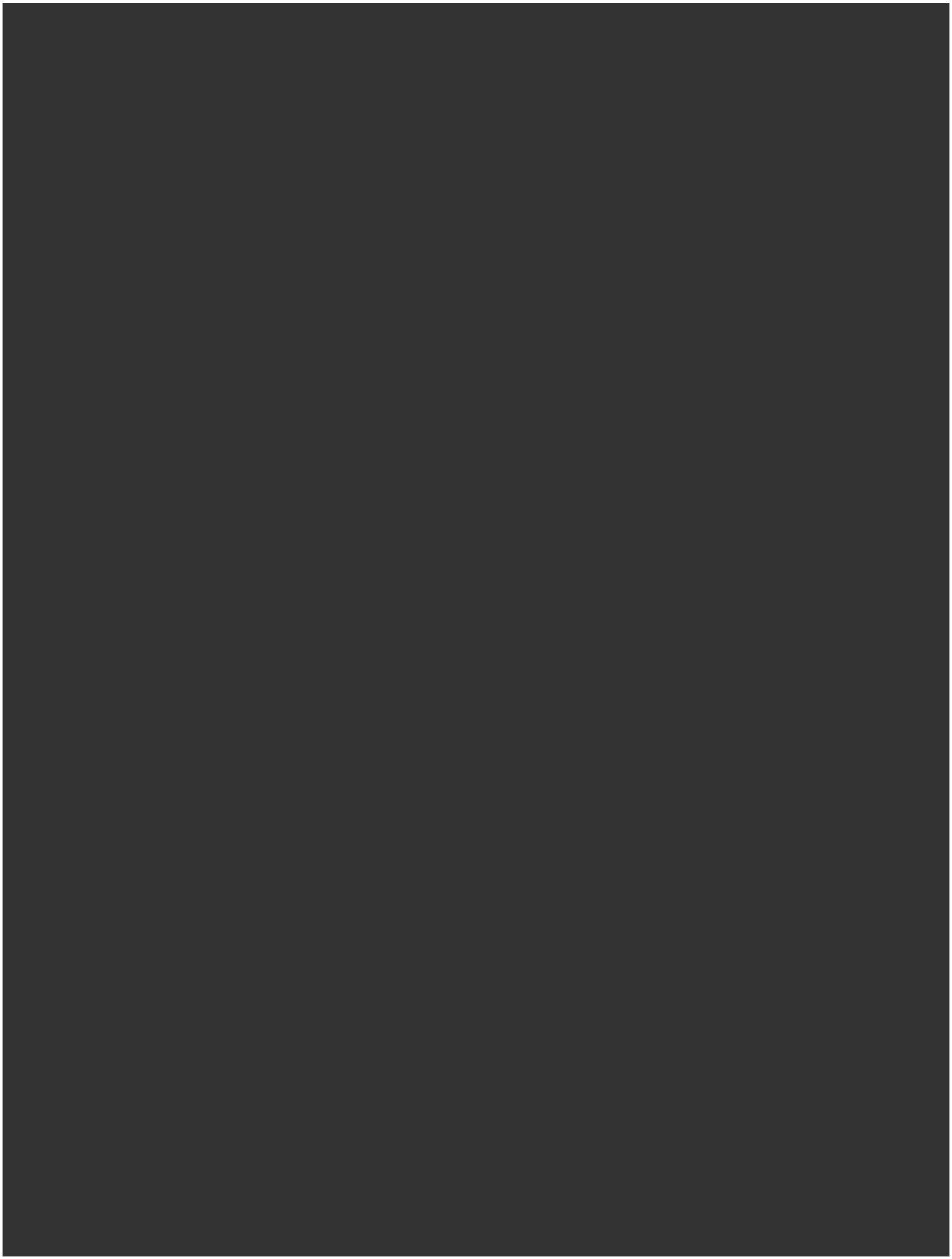
Therefore no unexplored branch can contain a better solution.

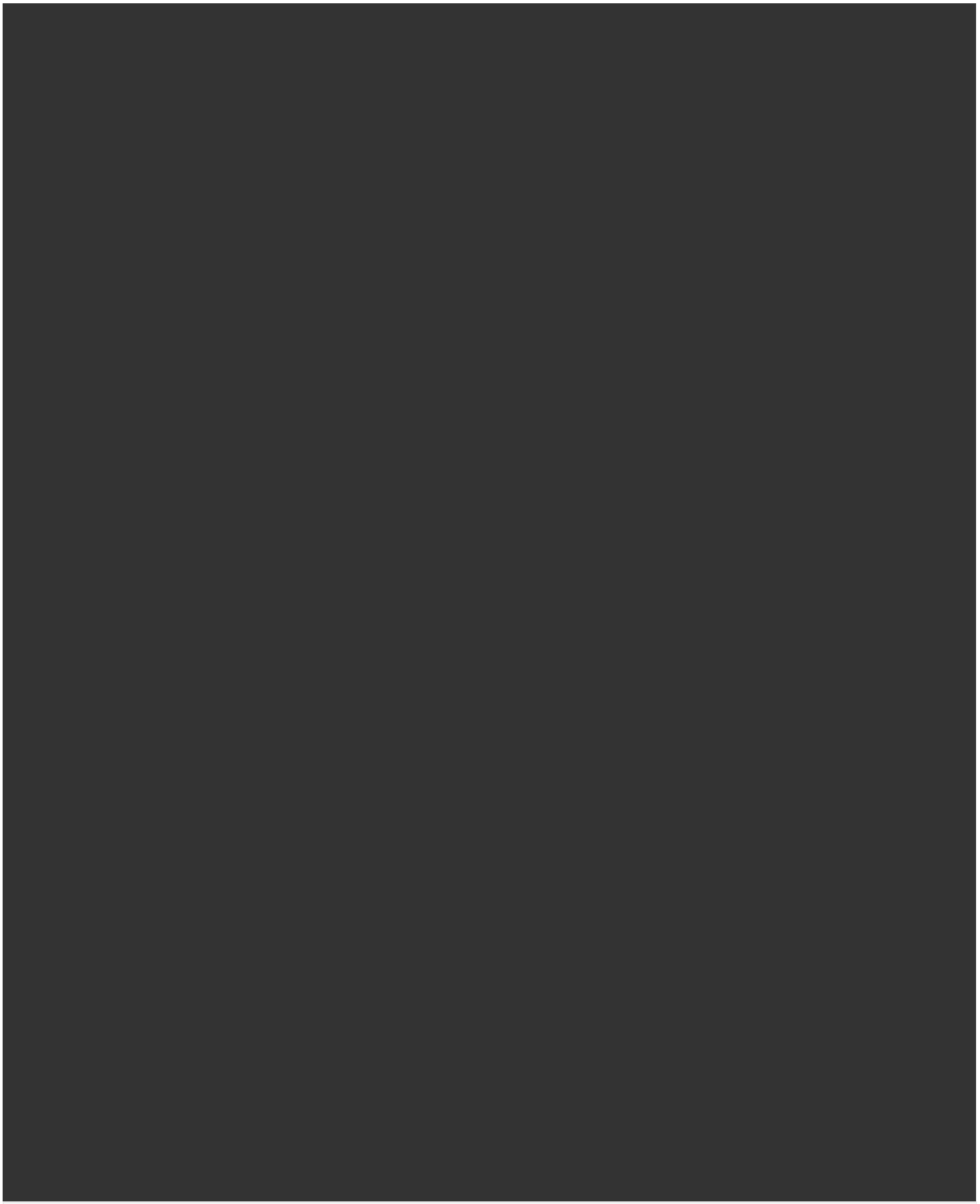
Can we stop at the first integer solution?	No. Here -4 improved to -6.
Why does a lower bound of -5.4 allow pruning?	Every integer descendant costs at least $-5.4>-6$.

Q & A


QUESTIONS
FOUND IN GOOGLE AUTOCOMPLETE
WHY ARE THERE SLAVES IN THE BIBLE
WHY ARE THERE ANTS IN MY LAPTOP
WHY ARE THERE GHOSTS
WHY ARE THERE SQUIRRELS
WHY ARE THERE DINOSAUR GHOSTS
WHY ARE THERE FEMALE MR MIMES
WHY ARE DOGS AFRAID OF FIREWORKS
WHY IS THERE NO KING IN ENGLAND
WHY AREN'T MY ARMS GROWING
WHY AREN'T THERE GUNS IN HARRY POTTER
WHY IS SEX SO IMPORTANT
WHY DO WHALES JUMP
WHY ARE WITCHES GREEN
WHY ARE THERE MIRRORS ABOVE BEDS
WHY DO I SAY UH
WHY IS SEA SALT BETTER
WHY ARE THERE TREES IN THE MIDDLE OF FIELDS
WHY IS THERE NOT A POKEMON MMO
WHY IS THERE LAUGHING IN TV SHOWS
WHY ARE THERE DOORS ON THE FREEWAY
WHY ARE THERE SO MANY SHOWSTOPPER RUNNING
WHY AREN'T THERE ANY COUNTRIES IN ANTARCTICA
WHY ARE THERE SCARY SOUNDS IN MINECRAFT
WHY IS THERE KICKING IN MY STOMACH
WHY ARE THERE TWO SUSHIES AFTER HTTP
WHY ARE THERE CELEBRITIES
WHY DO SNAKES EXIST
WHY DO OYSTERS HAVE PEARLS
WHY ARE DUCKS CALLED DUCKS
WHY DO THEY CALL IT THE CLAP
WHY ARE KYLE AND CARTMAN FRIENDS
WHY IS THERE AN ARROW ON AANG'S HEAD
WHY ARE TEXT MESSAGES BLUE
WHY ARE THERE MUSTACHES ON CLOTHES
WHY ARE THERE MUSTACHES ON CARS
WHY ARE THERE MUSTACHES EVERYWHERE
WHY ARE THERE SO MANY BIRDS IN OHIO
WHY IS THERE SO MUCH RAIN IN OHIO
WHY IS OHIO WEATHER SO WEIRD
WHY ARE THERE MALE AND FEMALE BIKES
WHY DO TESTICLES MOVE
WHY ARE THERE PSYCHICS
WHY ARE HATS SO EXPENSIVE
WHY IS THERE CAFFEINE IN MY SHAMPOO
WHY DO YOUR BOOBS HURT
WHY DO TWINS HAVE DIFFERENT FINGERPRINTS
WHY ARE AMERICANS AFRAID OF DRAGONS
WHY DO I SQUINT WHEN I SMILE
WHY DO SQUIRRELS DIE
WHY AREN'T ECONOMISTS RICH
WHY DO AMERICANS CALL IT SOCCER
WHY ARE MY EARS RINGING
WHY ARE THERE SO MANY AVENGERS
WHY ARE THE AVENGERS FIGHTING THE X MEN
WHY IS WOLVERINE NOT IN THE AVENGERS
WHY IS EARTH TILTED
WHY IS SPACE BLACK
WHY IS OUTER SPACE SO COLD
WHY ARE THERE PYRAMIDS ON THE MOON
WHY IS NASA SHUTTING DOWN
WHY ARE THERE TINY SPIDERS IN MY HOUSE
WHY DO SPIDERS COME INSIDE
WHY ARE THERE HUGE SPIDERS IN MY HOUSE
WHY ARE THERE LOTS OF SPIDERS IN MY HOUSE
WHY ARE THERE SPIDERS IN MY ROOM
WHY ARE THERE SO MANY SPIDERS IN MY ROOM
WHY DO SPIDER BITES ITCH
WHY IS DYING SO SCARY
WHY IS THERE NO GPS IN LAPTOPS
WHY DO KNEES CLICK
WHY AREN'T THERE E GRADES
WHY IS ISOLATION BAD
WHY DO BOYS LIKE ME
WHY DON'T BOYS LIKE ME
WHY IS THERE ALWAYS A JAVA UPDATE
WHY ARE THERE RED DOTS ON MY THIGHS
WHY IS LYING GOOD
WHY IS THERE A RED LINE THROUGH HTTPS ON FACEBOOK
WHY IS HTTPS CROSSED OUT IN RED
WHY IS THERE A LINE THROUGH HTTPS
WHY IS HTTPS IMPORTANT
WHY ARE THERE WEEDS
WHY DO I FEEL DIZZY
WHY ARE THERE SO MANY CROWS IN ROCHESTER, MN
WHY IS PSYCHIC WEAK TO BUG
WHY DO CHILDREN GET CANCER
WHY IS POSEIDON ANGRY WITH ODYSSEUS
WHY IS THERE ICE IN SPACE
WHY IS THERE AN OWL IN MY BACKYARD
WHY IS THERE AN OWL OUTSIDE MY WINDOW
WHY IS THERE AN OWL ON THE DOLLAR BILL
WHY DO OWLS ATTACK PEOPLE
WHY ARE AK 47s SO EXPENSIVE
WHY ARE THERE HELICOPTERS CIRCLING MY HOUSE
WHY ARE THERE GODS
WHY ARE THERE TWO SPOOKS
WHY IS MT VESUVIUS THERE
WHY DO THEY SAY T MINUS
WHY ARE THERE OBELISKS
WHY ARE WRESTLERS ALWAYS WET
WHY ARE OCEANS BECOMING MORE ACIDIC
WHY IS ARWEN DYING
WHY AREN'T MY QUAIL LAYING EGGS
WHY AREN'T MY QUAIL EGGS HATCHING
WHY AREN'T THERE ANY FOREIGN MILITARY BASES IN AMERICA
WHY ARE MY BOOBS ITCHY
WHY ARE CIGARETTES LEGAL
WHY ARE THERE DUCKS IN MY POOL
WHY IS JESUS WHITE
WHY IS THERE LIQUID IN MY EAR
WHY DO Q TIPS FEEL GOOD
WHY DO GOOD PEOPLE DIE
WHY IS LIFE SO BORING
WHY ARE ULTRASOUNDS IMPORTANT
WHY ARE ULTRASOUND PROBES DEFENSE
WHY IS STEALING WRONG

XKCD





Speaker notes

